

Perpendicular lines



1 Two lines that intersect each other at an angle of 90° .

2 -1

3

a No

b No

c Yes

d Yes

e No

f Yes

g Yes

h No

4

a $m = -\frac{36}{5}$

b $m = 10$

5

a $-\frac{1}{6}$

b $-\frac{1}{6}$

c -8

d 1

6

a $-\frac{8}{7}$

b $\frac{8}{7}$

c No

7

a $x = -8$

b $y = -8$

c $y = 2x + 6$

d $y = -\frac{x}{6} + 10$

e $y = -\frac{x}{6} - 4$

8

a $-\frac{1}{5}$

b $y = -\frac{x}{5} + 1$

9

a $\frac{-1}{10}$

b $(5, 10)$

c $x + 10y - 105 = 0$

Applications

10

a i 1

ii -1

b Yes, because their gradients multiply to -1 .

11



a

i $m = -1$

ii $m = 1$

iii $m = -1$

iv $m = 1$

b Square